Project -

Banking Analytics and Risk Prediction

A data-driven analysis of banking customers and financial products, with machine learning–based Risk Weighting prediction.

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Tools Used :

- **Python**
- **Sql**
- **Statistics**
- **PowerBi**
- **Machine Learning**

Problem Statement

ABC National Bank has experienced rapid growth in its customer base and financial products, including deposits, loans, credit cards, investments, and retirement savings. Although the bank collects large amounts of customer data, management lacks a consolidated view of customer behavior, profitability, and financial risk. Existing reports are largely static and do not provide the insights required for strategic decision-making.

The leadership team faces several challenges: identifying high-value customers, understanding which customer segments contribute the highest deposits and revenue, evaluating lending risk, measuring the effectiveness of loyalty programs, and assessing investment advisor performance. These gaps lead to missed cross-selling opportunities, inefficient resource allocation, and limited ability to personalize banking services.

The objective of this analytics project is to transform raw customer data into actionable business insights. The analysis will:

- Segment customers based on demographics and financial behavior
- Evaluate customer risk
- Identify profitable customer groups
- Measure advisor performance
- Explore relationships between income, deposits, loans, and other financial indicators

The final deliverables will include:

- A clean analytical dataset
- Statistical analysis
- An interactive Power BI dashboard
- Business recommendations

that help executives improve customer retention, optimize lending strategies, strengthen risk management, and increase overall customer lifetime value through data-driven decision-making

Dataset Describtion :

The Banking Dataset comprises four Excel sheets:

- Clients - Banking,
- Gender,
- Banking Relationship,
- Investment Advisor

combined into a single DataFrame that contains **3,000 records and 28 columns**. It encompasses various aspects such as customer demographics, income, savings, loans, deposits, financial assets, banking relationships, and risk-related information.

Data Quality

During the data validation process, no missing values or duplicate records were detected. Descriptive statistics were employed to assess the distribution and variability of numerical features.

	Age	Estimated Income	Superannuation Savings	Amount of Credit Cards	Credit Card Balance	Bank Loans	Bank Deposits	Checking Accounts	Saving Accounts	Foreign Currency Account	Business Lending	Properties Owned	Risk Weighting
count	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3.000000e+03	3.000000e+03	3.000000e+03	3.000000e+03	3000.000000	3.000000e+03	3000.000000	3000.000000
mean	51.039667	171305.034094	25531.599673	1.463667	3176.206780	5.913862e+05	5.715602e+05	3.210929e+05	2.329084e+05	29883.529774	8.667598e+05	1.518667	2.249333
std	19.854760	111935.808260	16259.950770	0.676387	2497.094724	4.575570e+05	5.457169e+05	2.820796e+05	2.300078e+05	23109.924007	6.412303e+05	1.102145	1.131191
min	17.000000	15919.475400	1482.030000	1.000000	1.174800	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	44.998400	0.000000e+00	0.000000	1.000000
25%	34.000000	82906.594300	12513.775000	1.000000	1236.633525	2.396281e+05	2.044004e+05	1.199475e+05	7.479441e+04	11916.546400	3.748251e+05	1.000000	1.000000
50%	51.000000	142313.479600	22357.355000	1.000000	2560.800750	4.797934e+05	4.633165e+05	2.428157e+05	1.640866e+05	24341.187200	7.113147e+05	2.000000	2.000000
75%	69.000000	242290.306650	35464.740000	2.000000	4522.633050	8.258130e+05	9.427546e+05	4.348749e+05	3.155750e+05	41966.390700	1.185110e+06	2.000000	3.000000
max	85.000000	522330.259200	75963.900000	3.000000	13991.994000	2.667557e+06	3.890598e+06	1.969923e+06	1.724118e+06	124704.866000	3.825962e+06	3.000000	5.000000

Feature Engineering

Several new features were developed, including:

- Joined Year
- Joined Month
- Total Assets
- Total Debt
- Net Financial Position
- Deposit-to-Income Ratio
- Loan-to-Income Ratio
- Customer Tenure (Years)

- Age Label

Additionally, a High Risk Percentage column was generated in Power BI for enhanced risk analysis and visualization.

Objective

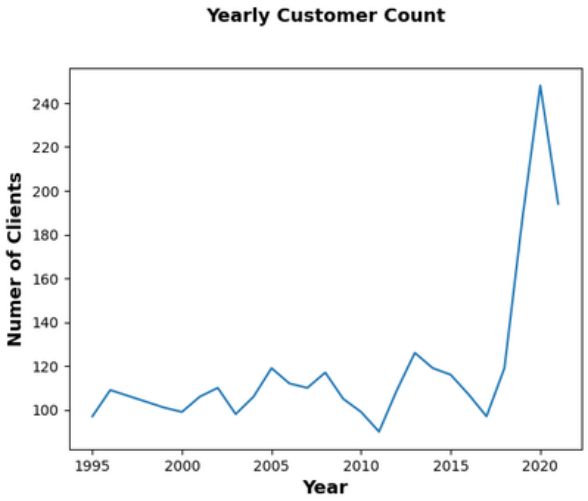
The goal is to extract actionable insights through exploratory data analysis (EDA), hypothesis testing, SQL analysis, risk prediction modeling, and interactive Power BI dashboards. This approach aims to facilitate data-driven decisions in customer segmentation, financial analysis, and risk management.

Python Analysis

EDA

1. Yearly Customer Acquisition Analysis

Purpose: To identify trends in customer acquisition over time and evaluate changes in the bank's customer growth.



Key Findings:

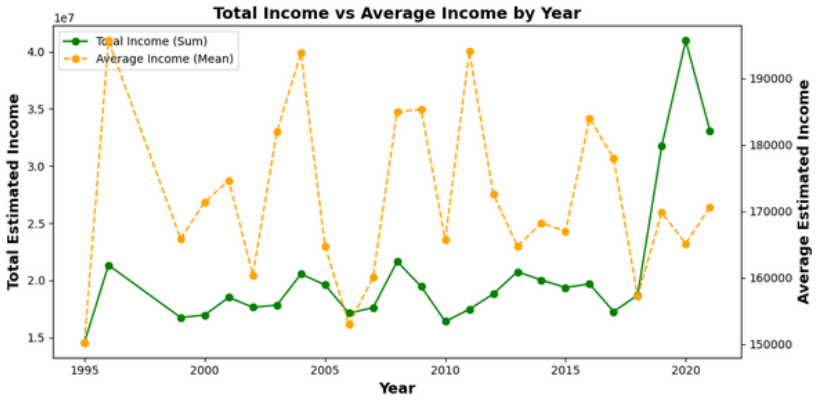
- Customer acquisition increased significantly during 2019–2020.
- Average annual customer acquisition rose from approximately 107 customers
- before 2019 to 209 per year from 2019 onward, an increase of around 95%.

.**Recommendations:**

- Investigate the factors behind the increase in customer acquisition.
- Identify successful acquisition channels and strategies to improve customer growth and retention.

2. Total Income vs. Average Income by Year

Purpose: To compare total estimated customer income with average income across joining years and understand changes in the financial profile of newly acquired customers.



Key Findings:

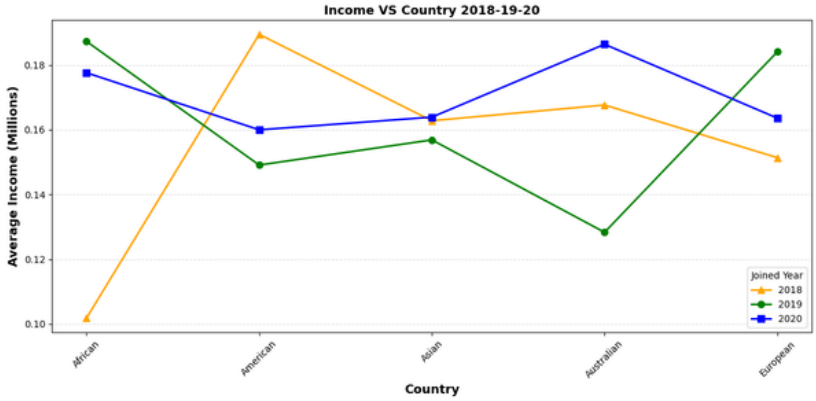
- Total estimated income increased substantially in 2019–2021.

Recommendations:

- Develop targeted products for customer segments with different income levels.

3. Average Income by Nationality Across Years (2018–2020)

Purpose: To compare average estimated income across nationalities and identify differences in the financial profiles of customers joining in different years.



Key Findings:

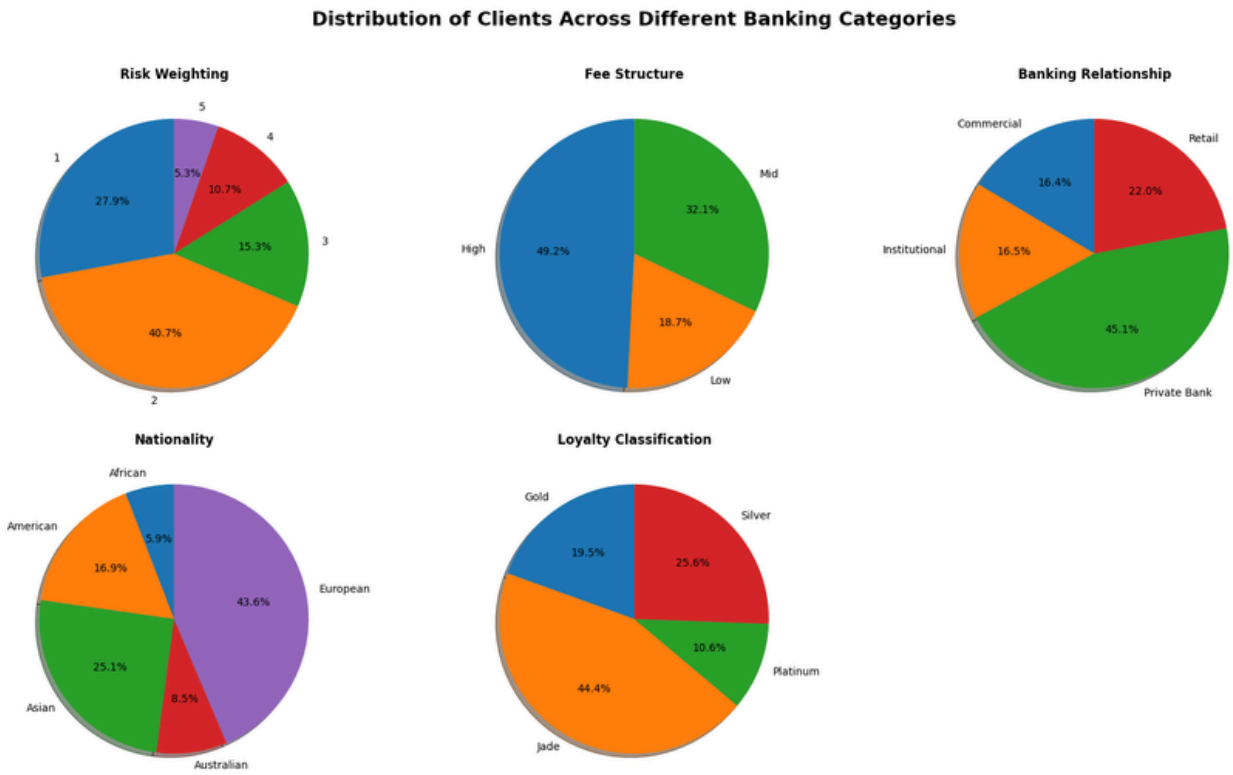
- American customers recorded relatively high average income in 2018, while Australian customers recorded the highest average in 2020.
- The changing patterns indicate differences in the income profiles of newly acquired customers.

Recommendations:

- Use nationality-level income patterns to guide customer segmentation and personalized product offerings.
- Validate these differences using customer counts and additional financial indicators before making market expansion decisions.

4. Customer Distribution Across Banking Categories

Purpose: To understand the composition of the customer base across risk weighting, fee structure, banking relationship, nationality, and loyalty classification.



Key Findings:

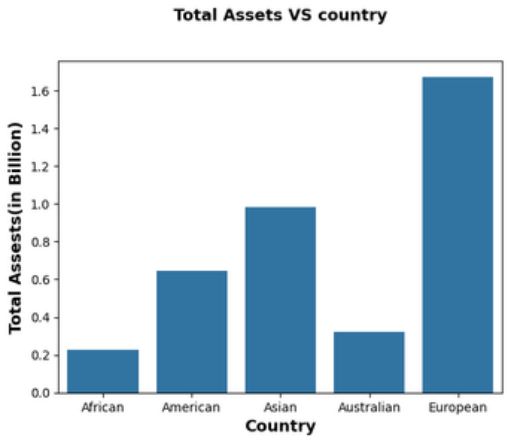
- Risk Weighting Category 2 represents the largest share at 40.7%.
- The High fee structure accounts for 49.2% of customers.
- Private Banking represents the largest banking relationship segment at 45.1%.
- European customers form the largest nationality group at 43.6%.
- Jade customers represent the largest loyalty segment at 44.4%**.**

Recommendations:

- Develop targeted retention and engagement strategies for major customer segments.
- Evaluate the profitability and service requirements of different banking relationships and fee structures.
- Examine the risk distribution within each customer segment to identify groups requiring closer monitoring.

5. Total Assets by Nationality

Purpose: To compare the total assets held by customers across nationalities and identify the segments contributing most to the bank's asset portfolio.



Key Findings:

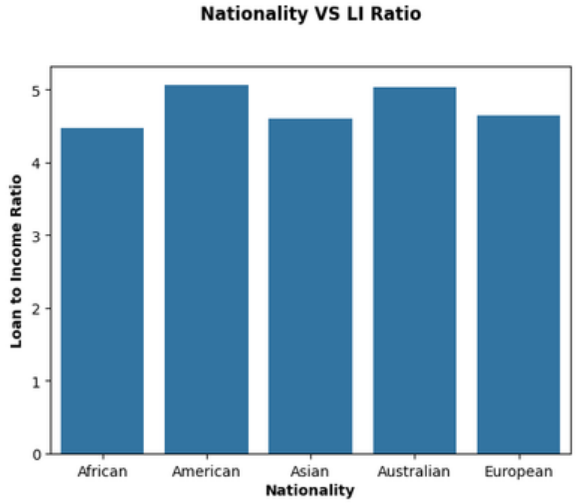
- European customers contribute the highest total assets.
- African customers contribute comparatively lower total assets.

Recommendations:

- Strengthen wealth-management and investment services for high-asset customer segments.
- Investigate opportunities to increase asset holdings through savings, investment, and deposit products.

6. Loan-to-Income Ratio by Nationality

Purpose: To compare borrowing relative to income across nationalities and identify customer groups that may require further credit-risk assessment.



Key Findings:

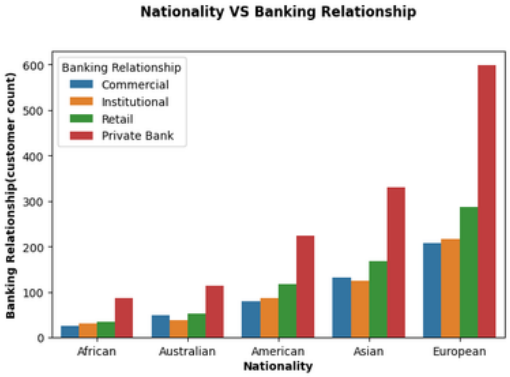
- American and Australian customers show the highest average loan-to-income ratios, at approximately 5.0.
- African customers have the lowest average ratio, at around 4.5.

Recommendations:

- Evaluate repayment capacity using credit history, debt obligations, and repayment behaviour before changing lending policies.
- Avoid treating nationality alone as a basis for credit decisions; use relevant individual financial and repayment indicators.

7. Banking Relationship by Nationality

Purpose: To examine how banking relationships are distributed across nationality groups and identify differences in customer preferences or relationship profiles.



Key Findings:

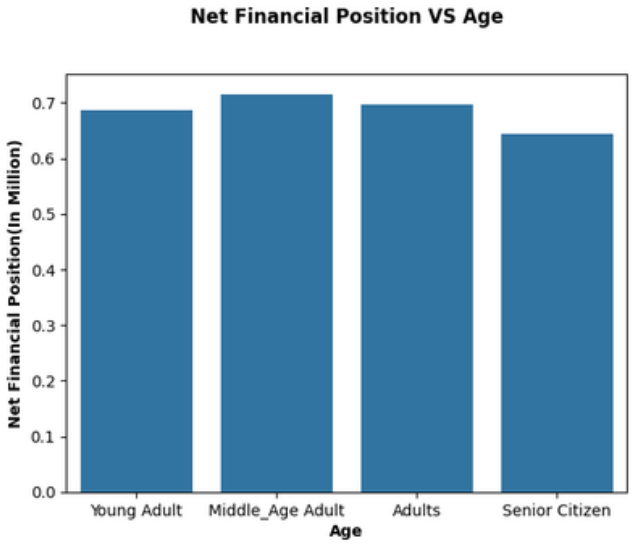
- Private Banking is the largest relationship category across the nationality groups shown.
- European customers account for the highest customer counts across the banking relationship categories.

Recommendations:

- Develop relationship-specific services to improve customer engagement and retention.
- Compare the profitability, product usage, and retention rates of Commercial, Institutional, Retail, and Private Banking customers.

8. Average Net Financial Position by Age Group

Purpose: To compare aggregate net financial positions across age groups and understand differences in customers' overall financial standing.



Key Findings:

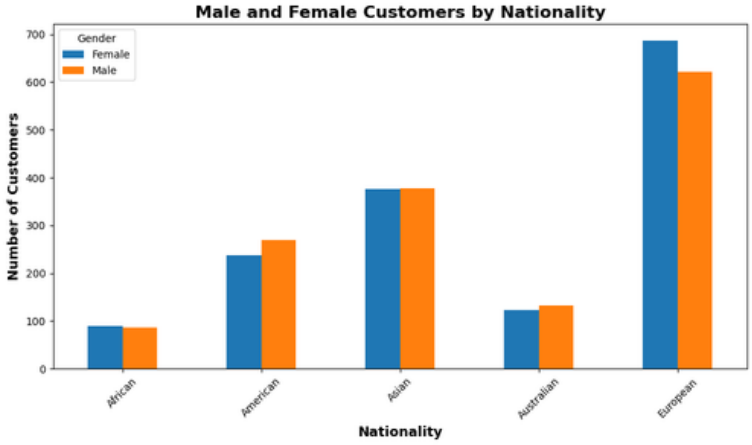
- Middle-aged adults and adults show the highest average net financial positions.
- Senior citizens have the lowest among the four groups.

Recommendations:

- Examine the underlying assets and liabilities of each age group to identify opportunities for improving financial product suitability.

9. Gender Distribution by Nationality

Purpose: To compare the number of male and female customers across nationalities and identify differences in the gender composition of the customer base.



Key Findings:

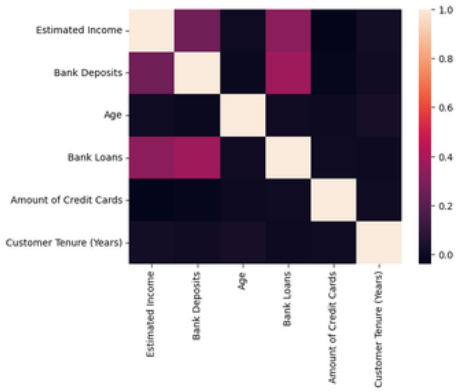
- European customers represent the largest customer group, with females outnumbering males.
- Asian customers show an almost equal distribution of male and female customers.

Recommendations:

- Investigate gender-based differences in banking product adoption, savings, investments, and loan usage.

10. Correlation Analysis of Financial Variables

Purpose: To measure the strength and direction of linear relationships among estimated income, bank deposits, age, bank loans, credit card count, and customer tenure.



Key Findings:

- Estimated income and bank deposits show a positive relationship.

Hypothesis Testing

1. Gender vs. Average Bank Deposits

Objective: Determine whether average bank deposits differ significantly between male and female customers.

Test selected: Welch's Independent Samples t-test. It compares the means of two independent groups without assuming equal population variances. The KDE plots show right-skewed distributions, so the test's robustness is worth considering alongside sample size and outliers.

Hypotheses

- (H0): Mean bank deposits are equal for male and female customers.
- (H_1): Mean bank deposits differ between male and female customers.

Results:

Metric	Result
t-statistic	-0.185
p-value	0.853
Significance level	0.05
Decision	Fail to reject H_0

Insight: No statistically significant difference in average bank deposits was detected between male and female customers. The observed difference is consistent with random sampling variation under the null hypothesis.

2. Gender vs. Bank Deposits — Mann–Whitney U Test

Objective: Check whether the deposit distributions differ between male and female customers using a non-parametric alternative.

Test selected: Mann–Whitney U test. It compares the relative ranks of two independent samples and does not require normally distributed data. It is particularly useful when distributions are skewed, although it is not automatically a test of medians.

Hypotheses :

- (H_0): The deposit distributions are the same for male and female customers.
- (H_1): The deposit distributions differ.

Results :

Metric	Result
U-statistic	1,116,917
p-value	0.736
Decision	Fail to reject H_0

Insight: The test found no statistically significant evidence of a difference in deposit distributions between the two gender groups. This supports the conclusion from the Welch t-test.

Recommendation: Since both tests reach the same conclusion, there is no strong statistical evidence in this dataset that gender is associated with deposit levels.

Note: *These two tests use the same deposit data, so they are complementary checks rather than two independent discoveries.*

3. Gender vs. Bank Loans

Objective: Determine whether average bank loan amounts differ significantly between male and female customers.

Test selected: Welch's Independent Samples t-test. It compares the average loan amounts of two independent groups while allowing their variances to differ.

Hypotheses

- (H_0): Mean bank loans are equal for male and female customers.
- (H_1): Mean bank loans differ between male and female customers.

Results

Metric	Result
t-statistic	1.432
p-value	0.152
Significance level	0.05
Decision	Fail to reject H_0

Insight: No statistically significant difference in average bank loans was detected between male and female customers. Although the sample means may differ, the evidence is insufficient to conclude that the population means differ.

4. Gender vs. Estimated Income

Objective: Assess whether average estimated income differs significantly between male and female customers.

Test selected: Welch's Independent Samples t-test, followed by Cohen's (d) to evaluate the practical magnitude of the difference. The t-test assesses statistical significance, while the effect size helps determine whether the difference is meaningful in practice.
Hypotheses

- (H_0): Mean estimated income is equal for male and female customers.
- (H_1): Mean estimated income differs between male and female customers.

Results

Metric	Result
t-statistic	1.978
p-value	0.048
Cohen's d	0.072
Decision	Reject H_0 at 5%

Insight: A statistically significant difference in estimated income was detected at the 5% significance level. However, Cohen's (d = 0.072) indicates a negligible standardized effect.

5. Banking Relationship vs. Average Bank Deposits

Objective: Determine whether average bank deposits differ across Retail, Commercial, Institutional, and Private Banking customers.

Tests selected:

1. Levene's test to assess the equal-variance assumption.
2. One-way ANOVA to compare mean deposits across four independent groups.

Hypotheses

- (H_0): All four banking relationship groups have equal mean deposits.
- (H_1): At least one group has a different mean deposit amount.

Result

Test	Statistic	p-value
Levene's test	0.464	0.707
One-way ANOVA (F)	0.603	0.613

Decision: Fail to reject H_0 at the 5% significance level.

Insight: The analysis did not detect a statistically significant difference in average deposits across the four banking relationship groups. Levene's test also found no evidence of unequal variances, supporting the equal-variance assumption.

Note: ANOVA tests whether group means differ; it does not establish that banking relationship has no effect. A non-significant result means the available evidence was insufficient to establish a difference.

6. Age Group vs. Risk Weighting

Objective: Investigate whether average Risk Weighting differs across Young Adult, Middle-Age Adult, Adult, and Senior Citizen groups.

Tests selected:

1. Levene's test to assess homogeneity of variance.
2. One-way ANOVA to compare mean Risk Weighting across the four age groups.

Hypotheses

- (H_0): Mean Risk Weighting is equal across all age groups.
- (H_1): At least one age group has a different mean Risk Weighting.

Results

Metric	Result
Chi-square statistic	11.187
Degrees of freedom	6
p-value	0.083
Decision	Fail to reject H_0 at 5%

Insight: No statistically significant difference in average Risk Weighting was detected across the four age groups. Levene's test found no significant evidence of unequal variances.

Important limitation: Risk Weighting is an ordinal score, so ANOVA treats it as approximately numeric and assumes meaningful differences between scores. As a robustness check, consider a Kruskal–Wallis test or analyse high-risk status as a binary outcome.

7. Age Group vs. Fee Structure

Objective: Determine whether the distribution of fee structures differs across age groups.

Test selected: Chi-square test of independence. Both Age Group and Fee Structure are categorical variables, making this test appropriate for assessing whether they are associated.

Hypotheses

- H_0): Age Group and Fee Structure are independent.
- (H_1): Age Group and Fee Structure are associated.

Results

Metric	Result
Chi-square statistic	11.187
Degrees of freedom	6
p-value	0.083
Decision	Fail to reject H_0 at 5%

Insight: The analysis did not find a statistically significant association between age group and fee structure at the 5% significance level. The observed differences in fee structure distributions could be due to random variation.

Risk Weighting Prediction Using ML

1. Objective

Develop an end-to-end machine learning pipeline to predict customer Risk Weighting (classes 1–5) using demographic, financial, and banking-related features. Three classification algorithms were evaluated to identify the best-performing model and understand the key factors associated with predicted risk categories

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Tools & Techniques: Python, Pandas, Scikit-learn, Logistic Regression, Random Forest, HistGradientBoosting, Stratified K-Fold Cross-Validation, RandomizedSearchCV, and classification evaluation metrics.

2. Data Preparation

The target variable contained five risk categories with an imbalanced distribution.

Risk WeightingCustomer Distribution

Risk Weighting	Customer Distribution
1	27.87%
2	40.73%
3	15.33%
4	10.73%
5	5.33%

Feature configuration: 22 input features comprising **18** numerical, **3** nominal categorical, and **1** ordinal categorical feature.
Numerical features included Estimated Income, Bank Loans, Bank Deposits, Total Assets, Total Debt, Net Financial Position, financial ratios, and Customer Tenure.
Categorical features included Gender, Banking Relationship, Nationality, and Fee Structure.
The dataset was divided into **2,400** training observations (80%) and **600** test observations (20%) using stratified sampling. This preserved approximately the same target-class proportions in both subsets.

3. Modeling Methodology

An end-to-end preprocessing and classification pipeline was developed to ensure consistent transformations and model evaluation.

Technique	Purpose
Stratified train-test split	Preserve class proportions in training and test data
Stratified 5-Fold CV	Evaluate model performance across multiple folds
RandomizedSearchCV	Identify effective hyperparameter combinations
Cross-validation scoring	Select models using macro F1-score
Classification report	Evaluate precision, recall, and F1-score for each class
Confusion matrix	Identify correct predictions and class misclassifications
Accuracy and macro F1-score	Assess overall and class-balanced performance

Model selection criterion: Macro F1-score was prioritized because it gives equal importance to each risk category, including underrepresented classes.

4. Model Comparison and Results

Three classification algorithms were trained and evaluated.

Model	Evaluation Method	Macro F1-score
Logistic Regression	5-fold CV	0.438 ± 0.019
Random Forest	Best 5-fold CV	**0.514**
HistGradientBoosting	Best 5-fold CV	0.502

Randomized hyperparameter search evaluated 40 configurations for Random Forest (200 cross-validation fits) and 12 configurations for HistGradientBoosting (60 fits).

Best Model: Random Forest Classifier
The Random Forest model achieved the highest cross-validated macro F1-score and was selected for final testing.

Final Test Metric	Result
Accuracy	**61.33%**
Macro F1-score	**49.11%**
Weighted F1-score	Approximately 61%

Class-wise Performance

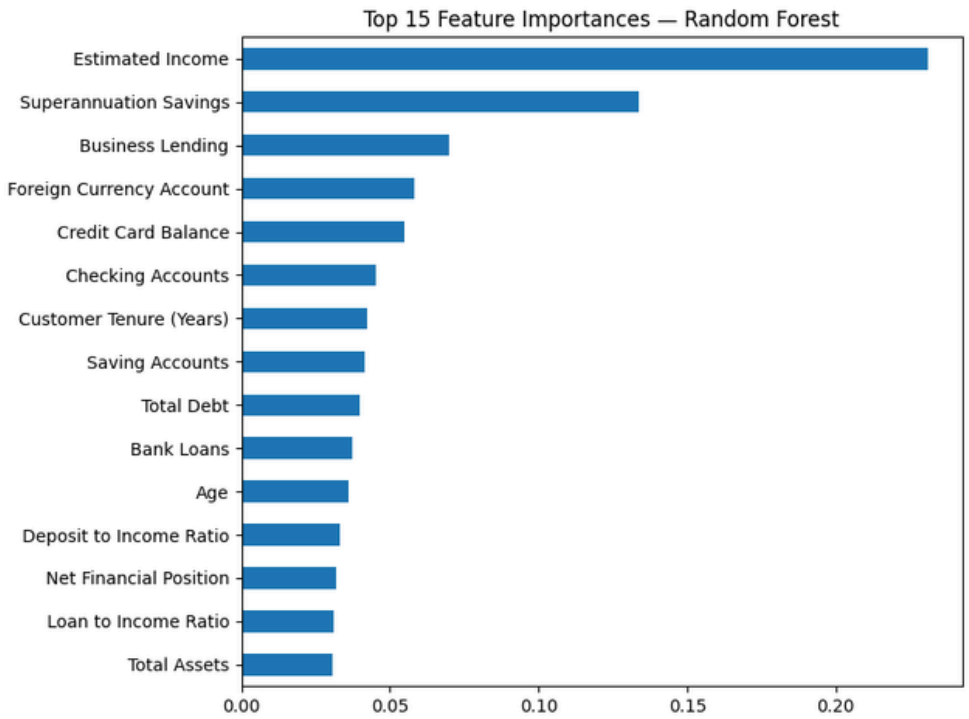
Risk Weighting	Precision	Recall	F1-score
1	0.78	0.79	0.78
2	0.67	0.69	0.68
3	0.41	0.49	0.44
4	0.30	0.23	0.26
5	0.41	0.22	0.29

Key findings:

- The model performed best for Risk Weighting classes 1 and 2.
- Recall was lower for classes 4 and 5, indicating difficulty identifying these less frequent categories.
- The **61.33%** accuracy should be interpreted alongside the **49.11%** macro F1-score, which reveals weaker performance across the full set of classes.

5. Important Feature Analysis

The Random Forest feature importances identified the variables most influential in its predictions.



SQL Business Analysis

1. Customer Segmentation and Wealth Analysis

Business question	Key result	Business insight
Which age group has the highest average deposits?	Middle-Age Adults: 0.691 million	This group has the highest average deposits among the four age groups.
Which age group has the highest total assets?	Middle-Age Adults: 1,001.64 million	This segment holds the largest aggregate assets in the displayed results.
Which occupations contribute the highest average income?	Environmental Tech: approximately 251,714	This occupation ranks highest in average income in the returned results.
Which occupation has the highest average net financial position?	Software Engineer IV: approximately 1.57 million	This occupation shows the strongest average net financial position among the displayed occupations.
Who are the highest-value customers by assets?	Top 10 customers ranked by average total assets	These customers can be examined for wealth-management, retention, and personalized service opportunities.

Key insight: Customer wealth is not distributed uniformly across age groups and occupations. Middle-Age Adults lead in aggregate assets and average deposits, while certain occupations show particularly high income or net financial position.

2. Customer Loyalty and Financial Behaviour

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Loyalty_Classification	Above_Avg_Tenure_Count	Total_Customers	Above_Avg_Tenure_Percentage
Silver	393	767	51.24
Platinum	159	317	50.16
Gold	285	585	48.72
Jade	619	1331	46.51

Analysis	Key result	Business insight
Loyalty vs. customer tenure	Silver: 51.24% above-average tenure	Silver has the highest proportion of customers whose tenure exceeds the overall average.
Loyalty vs. customer tenure	Platinum: 50.16%; Gold: 48.72%; Jade: 46.51%	Jade has the lowest above-average-tenure proportion among the four groups.
Age group vs. loan-to-income ratio	Adults: 4.87; Middle-Age Adults: 4.85	These two groups have the highest average ratios in the displayed results.

3. Banking Relationship Analysis

Banking relationship	Customers	Total assets (million)
Private Bank	1,352	1,759.44
Retail	660	818.82
Commercial	493	636.17
Institutional	495	628.51

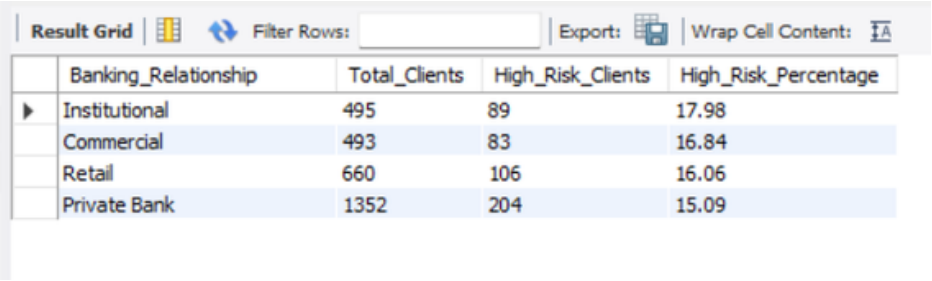
Purpose: Identify which banking relationship manages the largest customer base and asset portfolio.

Financial profile by banking relationship

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Banking_Relationship	customer_count	avg_income_million	avg_deposits_million	avg_bank_loan_million
Commercial	493	0.18	1.29	0.609
Institutional	495	0.176	1.27	0.563
Retail	660	0.168	1.241	0.577
Private Bank	1352	0.168	1.301	0.602

5. Risk Analysis and High-Risk Customer Identification

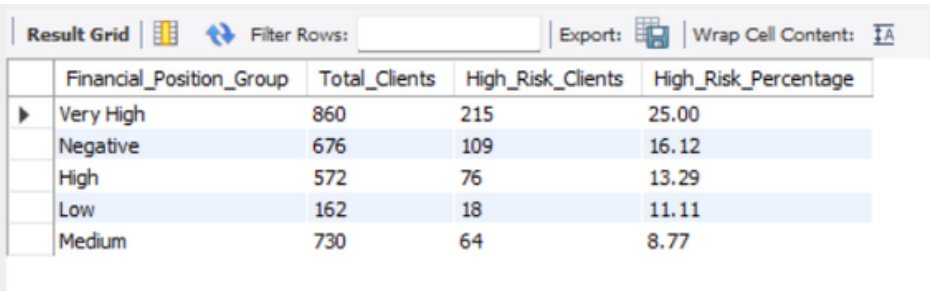


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Banking_Relationship	Total_Clients	High_Risk_Clients	High_Risk_Percentage
Institutional	495	89	17.98
Commercial	493	83	16.84
Retail	660	106	16.06
Private Bank	1352	204	15.09

Insight: Institutional Banking has the highest high-risk proportion at 17.98%, while Private Bank has the lowest at 15.09%. Private Bank nevertheless has the largest absolute number of high-risk customers (204), reflecting its larger customer base.

Net financial position vs. risk

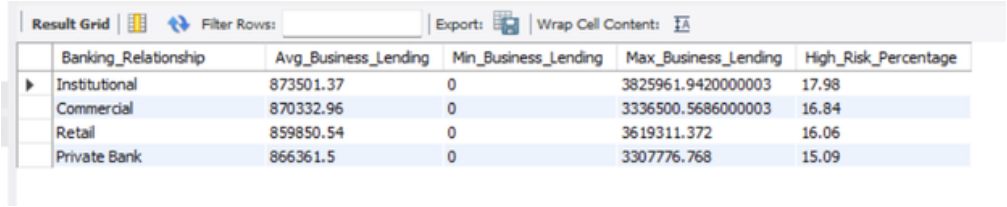


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Financial_Position_Group	Total_Clients	High_Risk_Clients	High_Risk_Percentage
Very High	860	215	25.00
Negative	676	109	16.12
High	572	76	13.29
Low	162	18	11.11
Medium	730	64	8.77

Key finding: The Very High financial-position group has the highest high-risk proportion, at 25%, while the Medium group has the lowest, at 8.77%.

Business Lending vs. Customer Risk



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Banking_Relationship	Avg_Business_Lending	Min_Business_Lending	Max_Business_Lending	High_Risk_Percentage
Institutional	873501.37	0	3825961.9420000003	17.98
Commercial	870332.96	0	3336500.5686000003	16.84
Retail	859850.54	0	3619311.372	16.06
Private Bank	866361.5	0	3307776.768	15.09

Key insights

1. Institutional Banking leads in both measures. It has the highest average business lending (873,501.37) and the highest high-risk proportion (17.98%).

2. Commercial Banking is a close second. Its average business lending is 870,332.96, with a high-risk proportion of 16.84%.

3. Private Bank has the lowest high-risk proportion. Despite average business lending of 866,361.50, its high-risk percentage is 15.09%

PowerBI Dashboard

Page 01 -



Page 02 -

